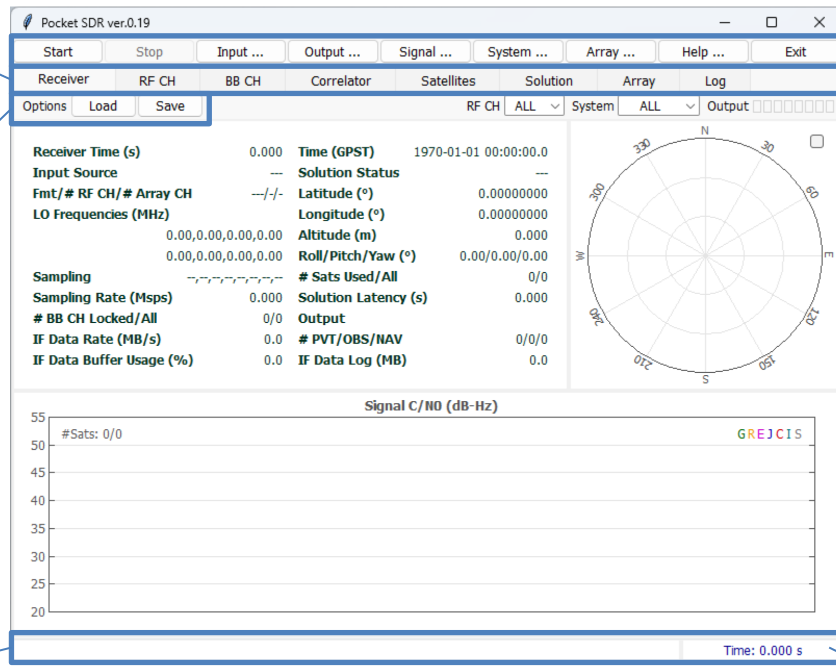
pocket_sdr.py help

ver. 0.19   2026-07-24

Initial Window

Page Tabs:
Tabs to switch current page

Load or Save Receiver Options:
The options format complies with Windows ini.



Command Buttons

Start	: Start the receiver
Stop	: Stop the receiver
Input...	: Show Input Options dialog
Output...	: Show Output Options dialog
Signal...	: Show Signal Options dialog
System...	: Show System Options dialog
Array...	: Show Array Options dialog
Help	: Show Help dialog
Exit	: Exit the receiver

Status Bar:
Status info or error message for the receiver

Receiver Time:
Elapsed time since the receiver started (s)

Receiver Page

Receiver Time:
Elapsed time since the receiver started
Input Source:
IF data source (RF_Frontend or IF_Data)
Fmt/# RF CH/# Array CH:
IF data format *3 and number of CHs
LO Frequencies (MHz):
Local oscillator frequencies for each RF CH (MHz)
Sampling:
IF data sampling type for each RF CH (I or IQ)
Sampling Rate (Msps):
IF data sampling rate
 # BB CH Locked/All:
Number of BB CHs tracked and all
IF Data Rate (MB/s):
IF data transfer rate
IF Data Buffer Usage (%):
Receiver internal IF data buffer usage rate
Time (GPST):
PVT Solution time in GPST
Solution Status:
PVT Solution status (- or FIX)
Latitude, Longitude and Altitude:
PVT Solution latitude/ longitude (deg) and ellipsoidal height (m)
Roll/Pitch/Yaw (deg):
Attitude solution by antenna array
 # Sats Used / All:
Number of used satellites for PVT and all satellites with OBS data
Solution Latency (s):
PVT Solution latency
 # PVT/# OBS/NAV Data:
Total count of PVT solutions, OBS and NAV data
IF Data Log (MB):
Recorded size of IF Data log

Receiver Status Panel

RF CH and System Selection for Skyplot and C/N0 Plot

Output Status Indicators
for stream 1-8 (PVT solution, OBS/NAV data, receiver log or IF data log)
Orange: waiting for connection
Green: data OK
Light-Green: data active
Red: error
White: no output

Satellite Positions in Skyplot *1 *2

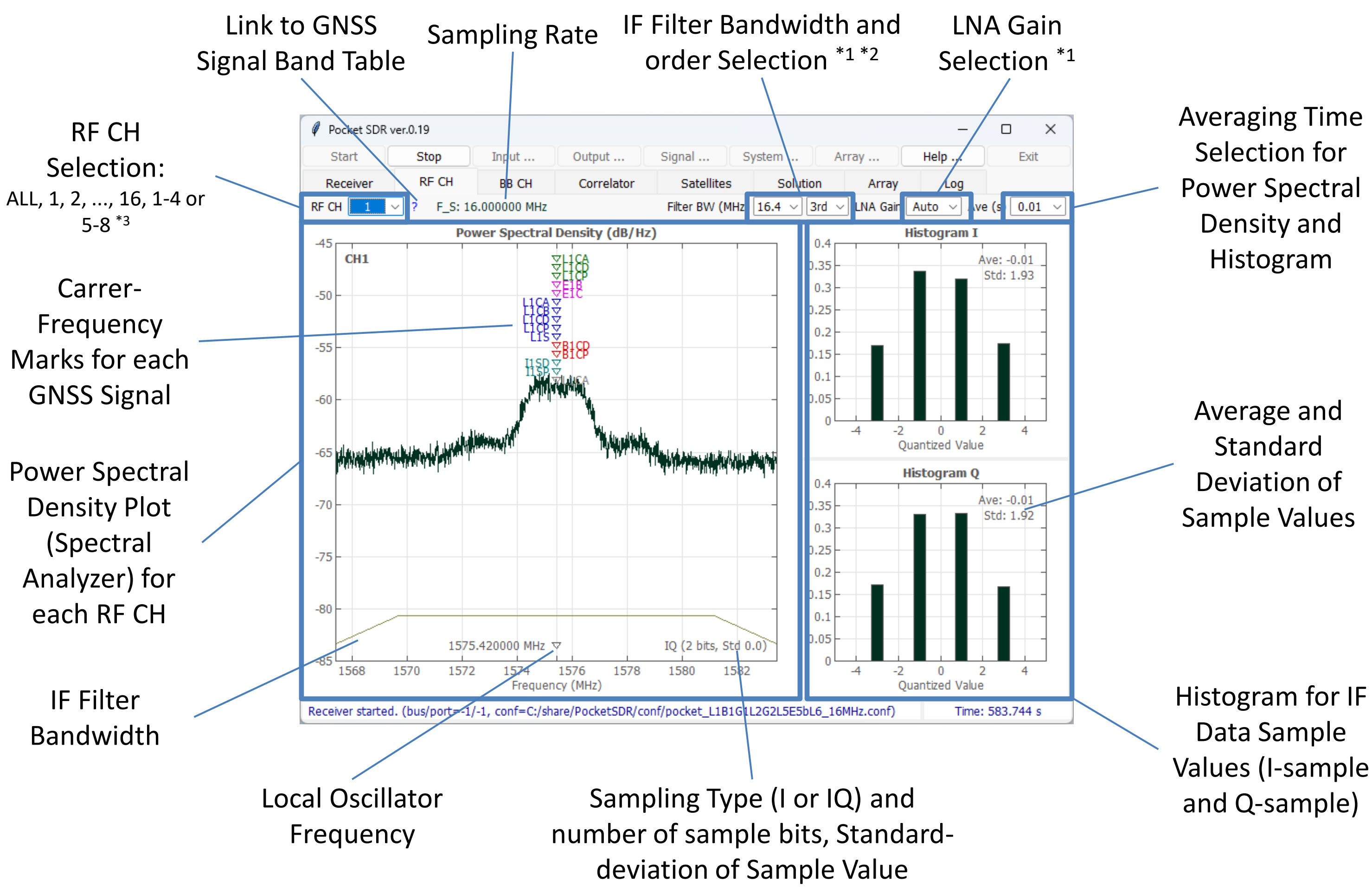
Number of used satellites and all satellites with OBS data

Signal C/N0 (dB-Hz) Plot for Each Satellite and Signals (stacked) *1 *2

* 1 GNSS System Color:
Green: GPS
Orange: GLONASS
Magenta: Galileo
Blue: QZSS
Red: BeiDou
Blue-Green: NavIC
Grey: SBAS
White or light-color: unused satellites for solution

*2 Satellite ID complies with RINEX convention
*3 INT8 (int8 I-sampling), INT8X2 (interleaved-int8 IQ-sampling), RAW8, RAW16, RAW32 (Pocket SDR FE), CS8 or CS16 (SoapySDR devices)

RF CH Page



*1 Only valid for RF_Frontend Input, *2 Only valid for IQ-sampling
*3 RF CH 9, 10, ..., 16 are beamforming CHs by antenna array

BB CH Page

RF CH Selection:
ALL, 1, 2, ..., 16

GNSS System Selection:
ALL, GPS, GLONASS, ...

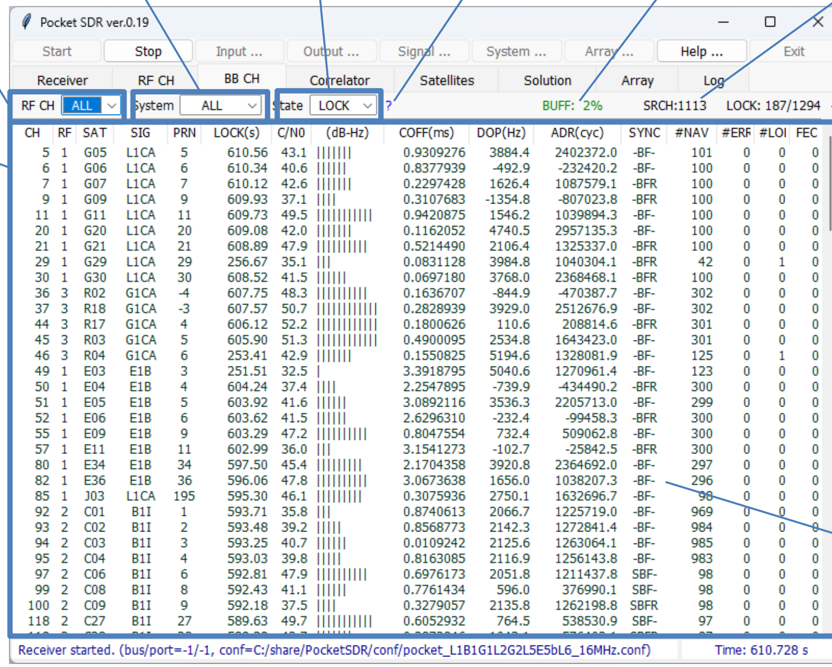
BB CH State Selection:
LOCK or ALL

Link to GNSS Signal Table

Internal IF Data Buffer Usage Rate in %

BB CH No Searching Signal

Number of Locked and All BB CHs



CH	RF	SAT	SIG	PRN	LOCK(s)	C/N0	(dB-Hz)	COFF(ms)	DOP(Hz)	ADR(cyc)	SYNC	#NAV	#ERR	#LOL	FEC
5	1	G05	L1CA	5	610.56	43.1	|||||||	0.9309276	3884.4	2402372.0	-BF-	101	0	0	0
6	1	G06	L1CA	6	610.34	40.6	||||||	0.8377939	-492.9	-232420.2	-BF-	100	0	0	0
7	1	G07	L1CA	7	610.12	42.6	|||||||	0.2297428	1626.4	1087579.1	-BFR	100	0	0	0
9	1	G09	L1CA	9	609.93	37.1	||||	0.3107683	-1354.8	-807023.8	-BFR	100	0	0	0
11	1	G11	L1CA	11	609.73	49.5	|||||||||||	0.9420875	1546.2	1039894.3	-BF-	100	0	0	0
20	1	G20	L1CA	20	609.08	42.0	|||||||	0.1162052	4740.5	2957135.3	-BF-	100	0	0	0
21	1	G21	L1CA	21	608.89	47.9	||||||||||	0.5214490	2106.4	1325337.0	-BFR	100	0	0	0
29	1	G29	L1CA	29	256.67	35.1	|||	0.0831128	3984.8	1040304.1	-BFR	42	0	1	0
30	1	G30	L1CA	30	608.52	41.5	||||||	0.0697180	3768.0	2368468.1	-BFR	100	0	0	0
36	3	R02	G1CA	-4	607.75	48.3	||||||||||	0.1636707	-844.9	-470387.7	-BF-	302	0	0	0
37	3	R18	G1CA	-3	607.57	50.7	||||||||||||	0.2828939	3929.0	2512676.9	-BF-	302	0	0	0
44	3	R17	G1CA	4	606.12	52.2	||||||||||||	0.1800626	110.6	208814.6	-BFR	301	0	0	0
45	3	R03	G1CA	5	605.90	51.3	||||||||||||	0.4900095	2534.8	1643423.0	-BF-	301	0	0	0
46	3	R04	G1CA	6	253.41	42.9	|||||||	0.1550825	5194.6	1328081.9	-BF-	125	0	1	0
49	1	E03	E1B	3	251.51	32.5	|	3.3918795	5040.6	1270961.4	-BF-	123	0	0	0
50	1	E04	E1B	4	604.24	37.4	||||	2.2547895	-739.9	-434490.2	-BFR	300	0	0	0
51	1	E05	E1B	5	603.92	41.6	||||||	3.0892116	3536.3	2205713.0	-BF-	299	0	0	0
52	1	E06	E1B	6	603.62	41.5	||||||	2.6296310	-232.4	-99458.3	-BFR	300	0	0	0
55	1	E09	E1B	9	603.29	47.2	||||||||||	0.8047554	732.4	509062.8	-BF-	300	0	0	0
57	1	E11	E1B	11	602.99	36.0	|||	3.1541273	-102.7	-25842.5	-BFR	300	0	0	0
80	1	E34	E1B	34	597.50	45.4	|||||||||	2.1704358	3920.8	2364692.0	-BF-	297	0	0	0
82	1	E36	E1B	36	596.06	47.8	||||||||||	3.0673638	1656.0	1038207.3	-BF-	296	0	0	0
85	1	J03	L1CA	195	595.30	46.1	|||||||||	0.3075936	2750.1	1632696.7	-BF-	98	0	0	0
92	2	C01	B1I	1	593.71	35.8	|||	0.8740613	2066.7	1225719.0	-BF-	969	0	0	0
93	2	C02	B1I	2	593.48	39.2	|||||	0.8568773	2142.3	1272841.4	-BF-	984	0	0	0
94	2	C03	B1I	3	593.25	40.7	||||||	0.0109242	2125.6	1263064.1	-BF-	985	0	0	0
95	2	C04	B1I	4	593.03	39.8	|||||	0.8163085	2116.9	1256143.8	-BF-	983	0	0	0
97	2	C06	B1I	6	592.81	47.9	||||||||||	0.6976173	2051.8	1211437.8	SBF-	98	0	0	0
99	2	C08	B1I	8	592.43	41.1	||||||	0.7761434	596.0	376990.1	SBF-	98	0	0	0
100	2	C09	B1I	9	592.18	37.5	||||	0.3279057	2135.8	1262198.8	SBFR	98	0	0	0
118	2	C27	B1I	27	589.63	49.7	|||||||||||	0.6052932	764.5	538530.9	SBF-	97	0	0	0

Status for each BB CH:

CH	: BB CH no	
RF	: Assigned RF CH no for BB CH	
SAT	: Satellite ID	
SIG	: Signal ID	
PRN	: PRN number or FCN for GLONASS FDMA signal	
LOCK	: Continuous lock time (s)	
C/N0	: Signal C/N0 (dB-Hz)	
(dB-Hz)	: Signal C/N0 Bar (20-50 dB-Hz)	
COFF	: Code offset (ms)	
DOP	: Doppler frequency (Hz)	
ADR	: Accumulated Doppler range (cycle)	
SYNC	: Synchronization status	
#NAV	: Decoded subframe or message count	
#ERR	: NAV data error count	
#LOL	: Loss-of-lock count	
FEC	: Number of error bits corrected by FEC (-1: correction unable)	

Synchronization Status:

S : Secondary code sync
B : NAV symbol (bit) sync
F : Subframe or message sync
R : Polarity of subframe or message sync (R: reversed)

Correlator Page

BB CH Selection *1: 1, 2, 3, ...

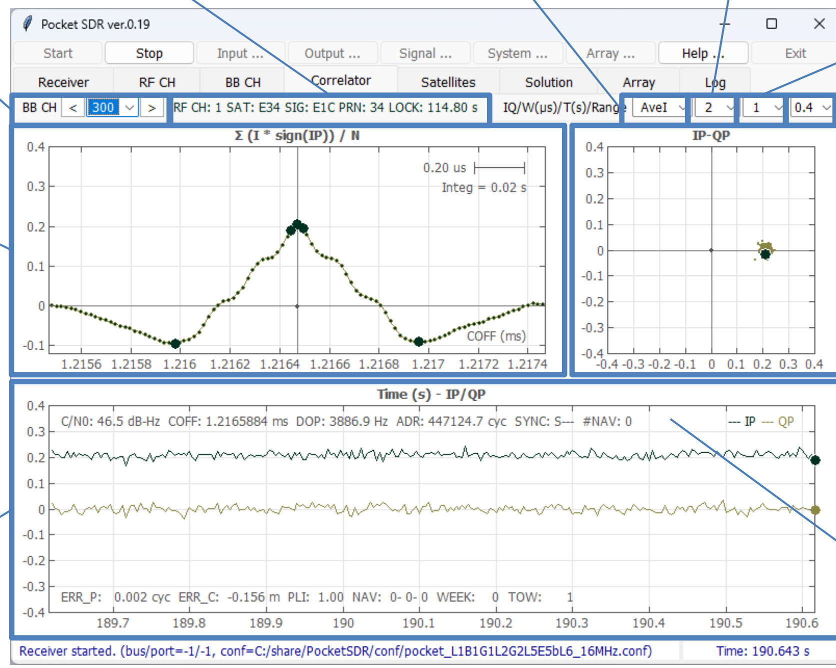
RF CH, Satellite ID, Signal ID PRN No. and Lock Time of BB CH

Correlation Plot Type Selection

X-axis Range Selection for Correlation Plot

Time Span Selection for Correlation History or Constellation Plot

Y-axis Range Selection



Correlation Plot

X-axis: code offset (ms)
Y-axis: correlations
(I * sign(IP), sqrt($I^2 + Q^2$) or sqrt(sum($I^2 + Q^2$))) *2
Large dots:
correlator positions for VE, E, P, L and VL *3

Constellation Plot

X-axis: IP-correlation
Y-axis: QP-correlation
Large dot:
current correlation

Correlation History Plot

X-axis: receiver time (s)
Y-axis: IP or QP-correlation

C/N0, code offset, Doppler Frequency, accumulated Doppler range, synchronization status, decoded subframe or message count

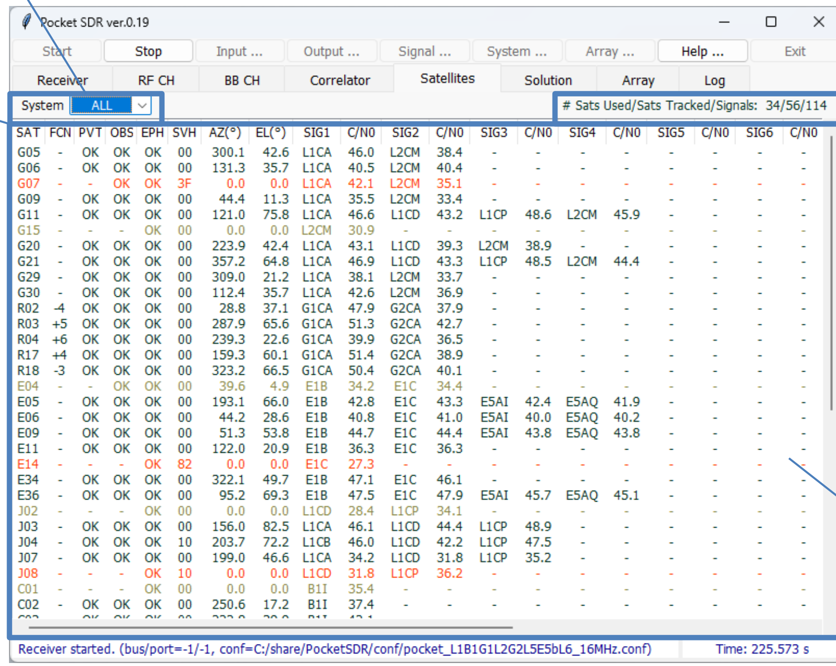
*1 BB CH can be selected by clicking BB CH Status row in BB CH Page
*2 Integration time is based on non-coherent integration time for DLL
*3 VE and VL are only shown in case of BOC-modulation and Bump-Jump enabled

Satellites Page

System Selection

Status for each Satellite:

SAT : Satellite ID
FCN : Frequency channel no (FCN) for GLONASS FDMA signals
PVT : Satellite used in PVT solution or not
OBS : Satellite L1 OBS data availability
EPH : Satellite ephemeris availability
SVH : Satellite health flag in ephemeris (HEX)
AZ : Azimuth angle (deg)
EL : Elevation angle (deg)
SIG1 : Signal 1 ID
C/N0 : C/N0 of signal 1 (dB-Hz)
SIG2 : Signal 2 ID
C/N0 : C/N0 for signal 2 (dB-Hz)
SIG3 : Signal 3 ID
C/N0 : C/N0 for signal 3 (dB-Hz)
... ...



SAT	FCN	PVT	OBS	EPH	SVH	AZ(°)	EL(°)	SIG1	C/N0	SIG2	C/N0	SIG3	C/N0	SIG4	C/N0	SIG5	C/N0	SIG6	C/N0
G05	-	OK	OK	OK	00	300.1	42.6	L1CA	46.0	L2CM	38.4	-	-	-	-	-	-	-	-
G06	-	OK	OK	OK	00	131.3	35.7	L1CA	40.5	L2CM	40.4	-	-	-	-	-	-	-	-
G07	-	-	OK	OK	3F	0.0	0.0	L1CA	42.1	L2CM	35.1	-	-	-	-	-	-	-	-
G09	-	OK	OK	OK	00	44.4	11.3	L1CA	35.5	L2CM	33.4	-	-	-	-	-	-	-	-
G11	-	OK	OK	OK	00	121.0	75.8	L1CA	46.6	L1CD	43.2	L1CP	48.6	L2CM	45.9	-	-	-	-
G15	-	-	-	OK	00	0.0	0.0	L2CM	30.9	-	-	-	-	-	-	-	-	-	-
G20	-	OK	OK	OK	00	223.9	42.4	L1CA	43.1	L1CD	39.3	L2CM	38.9	-	-	-	-	-	-
G21	-	OK	OK	OK	00	357.2	64.8	L1CA	46.9	L1CD	43.3	L1CP	48.5	L2CM	44.4	-	-	-	-
G29	-	OK	OK	OK	00	309.0	21.2	L1CA	38.1	L2CM	33.7	-	-	-	-	-	-	-	-
G30	-	OK	OK	OK	00	112.4	35.7	L1CA	42.6	L2CM	36.9	-	-	-	-	-	-	-	-
R02	-4	OK	OK	OK	00	28.8	37.1	G1CA	47.9	G2CA	37.9	-	-	-	-	-	-	-	-
R03	+5	OK	OK	OK	00	287.9	65.6	G1CA	51.3	G2CA	42.7	-	-	-	-	-	-	-	-
R04	+6	OK	OK	OK	00	239.3	22.6	G1CA	39.9	G2CA	36.5	-	-	-	-	-	-	-	-
R17	+4	OK	OK	OK	00	159.3	60.1	G1CA	51.4	G2CA	38.9	-	-	-	-	-	-	-	-
R18	-3	OK	OK	OK	00	323.2	66.5	G1CA	50.4	G2CA	40.1	-	-	-	-	-	-	-	-
E04	-	-	OK	OK	00	39.6	4.9	E1B	34.2	E1C	34.4	-	-	-	-	-	-	-	-
E05	-	OK	OK	OK	00	193.1	66.0	E1B	42.8	E1C	43.3	E5AI	42.4	E5AQ	41.9	-	-	-	-
E06	-	OK	OK	OK	00	44.2	28.6	E1B	40.8	E1C	41.0	E5AI	40.0	E5AQ	40.2	-	-	-	-
E09	-	OK	OK	OK	00	51.3	53.8	E1B	44.7	E1C	44.4	E5AI	43.8	E5AQ	43.8	-	-	-	-
E11	-	OK	OK	OK	00	122.0	20.9	E1B	36.3	E1C	36.3	-	-	-	-	-	-	-	-
E14	-	-	-	OK	82	0.0	0.0	E1C	27.3	-	-	-	-	-	-	-	-	-	-
E34	-	OK	OK	OK	00	322.1	49.7	E1B	47.1	E1C	46.1	-	-	-	-	-	-	-	-
E36	-	OK	OK	OK	00	95.2	69.3	E1B	47.5	E1C	47.9	E5AI	45.7	E5AQ	45.1	-	-	-	-
J02	-	-	-	OK	00	0.0	0.0	L1CD	28.4	L1CP	34.1	-	-	-	-	-	-	-	-
J03	-	OK	OK	OK	00	156.0	82.5	L1CA	46.1	L1CD	44.4	L1CP	48.9	-	-	-	-	-	-
J04	-	OK	OK	OK	10	203.7	72.2	L1CB	46.0	L1CD	42.2	L1CP	47.5	-	-	-	-	-	-
J07	-	OK	OK	OK	00	199.0	46.6	L1CA	34.2	L1CD	31.8	L1CP	35.2	-	-	-	-	-	-
J08	-	-	-	OK	10	0.0	0.0	L1CD	31.8	L1CP	36.2	-	-	-	-	-	-	-	-
C01	-	-	-	OK	00	0.0	0.0	B1I	35.4	-	-	-	-	-	-	-	-	-	-
C02	-	OK	OK	OK	00	250.6	17.2	B1I	37.4	-	-	-	-	-	-	-	-	-	-

Number of satellites used for PVT solution, number of satellites tracked, number of all tracked signals

Red: unhealthy satellite
Grey: unused satellite for PVT solution

Solution Page

PVT Solution Plot Selection:
Pos ENU or Pos Horiz

Show Ref Pos Dialog Button

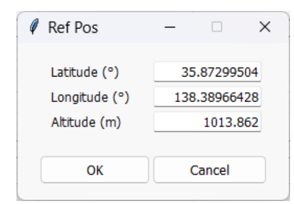
PVT Solution Clear Button

Time Span and Y-axis Range Selections

Current Solution and Number of Used/All Satellites

PVT Solution Plot:
Pos E, N, U: position errors east-west, north-south, up-down *1,
Sats: Number of used/all satellites
or
Position error horizontal *1 (east-west and north-west, horizontal scatter plot)

Ref Pos Dialog



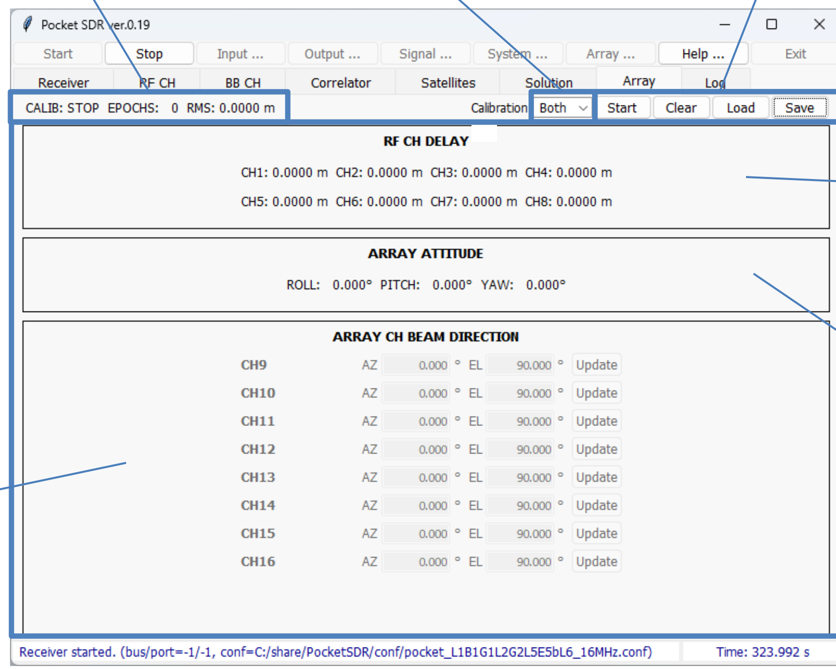
*1 The reference position is automatically selected from the first epoch solution.
It can be modified by using Ref Pos Dialog

Array Page

Array Calibration Status:
STOP or RUN, Total epochs for calibration and RMS residuals

Array Calibration Targets:
Both: Dealy and Attiude
Delay: Delay only
Att: Attitude only

Array Calibration Operation Buttons:
Start or Stop, Clear, Load and Save



RF CH DELAY:
RF CH delay calibration results referenced to RF CH1

ARRAY ATTITUDE:
Array attitude calibration results (ROLL, PITCH, YAW angles)

ARRAY CH BEAM DIRECTION:
Beam direction control for each Array CH. Set AZ (azimuth) and EL (elevation) angles and push Update Button.

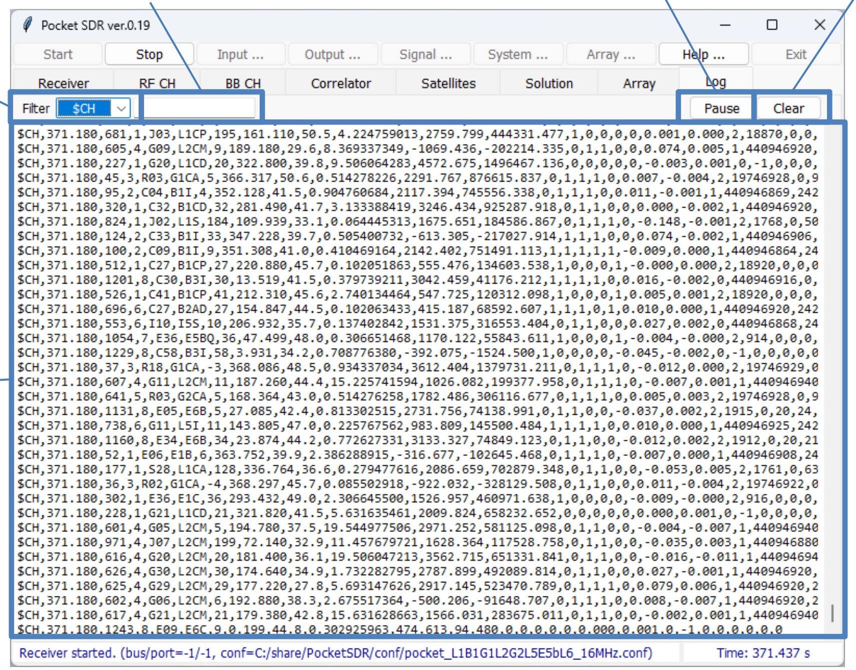
Log Page

Additional Receiver Log Filter:
filter string, like G01 or L1CA, separated by spaces, | indicates OR condition (e.g., G10|G02|G03)

Receiver Log Pause Button

Receiver Log Clear Button

Receiver Log Type Selection:
Blank: all log



Receiver Logs *1 *2

*1 Refer p.16 for the receiver log formats

*2 To copy the logs to Clip-board, push Pause button, select the logs and push Ctrl-C of the keyboard

Input Options Dialog

Input Source Selection:

RF Frontend or IF Data

Device Configuration File:

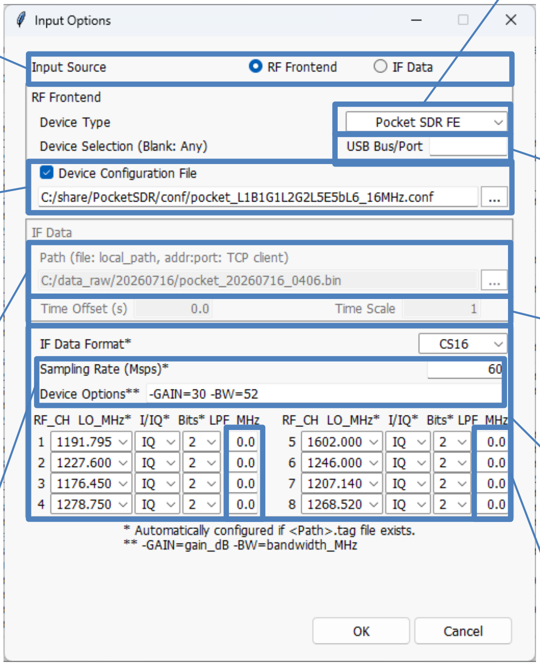
RF Frontend configuration settings for Pocket SDR FE (MAX2771 register) to be loaded at the receiver started. Check Device Configuration File and fill the file path. If unchecked, the current configuration for the device is used. Ignored for other devices.

IF Data File [*1]:

IF data file path, if IF Data selected as the input source. The IF data can be captured by pocket_dump or as IF data log of the receiver with tag files.

Sampling Rate and Device Options for Soapy SDR Devices:

-GAIN=gain (LNA gain),
-BW=bw (Filter Bandwidth),
Options are separated by spaces.



RF Frontend Type Selection:

Pocket SDR FE: Pocket SDR FE 2CH, 4CH, 8CH
USRP (uhd): USRP via Soapy SDR
LimeSDR (lime): LimeSDR via Soapy
LimeSDR (limesuiteng): LimeSDR via newer driver with Soapy
BladeRF (bladrf): bladeRF via Soapy
RTL-SDR (rtlsdr): RTL-SDR via Soapy
Plute-SDR (plutosdr): PlutoSDR via Soapy
Airspy (airspy): Airspy via Soapy

USB Bus/Port:

To select Pocket SDR FE in case of multiple devices connected. If blank, the device is automatically selected.

Time Offset and Scale:

Start time offset and time scale to playback the IF Data.

IF Data File Property [*1]:

IF Data Format, Sampling Rate, LO frequency, Sampling Type (I or IQ), Number of sample bits for each RF CH. If the tag file exist for the IF data file, these are automatically configured according to the tag file.

Input LPF Settings:

Set two-sided bandwidth (MHz) of LPF for each RF CH. 0 indicates disabled.

*1 Refer to p.15 for the IF data file formats

Output Options Dialog

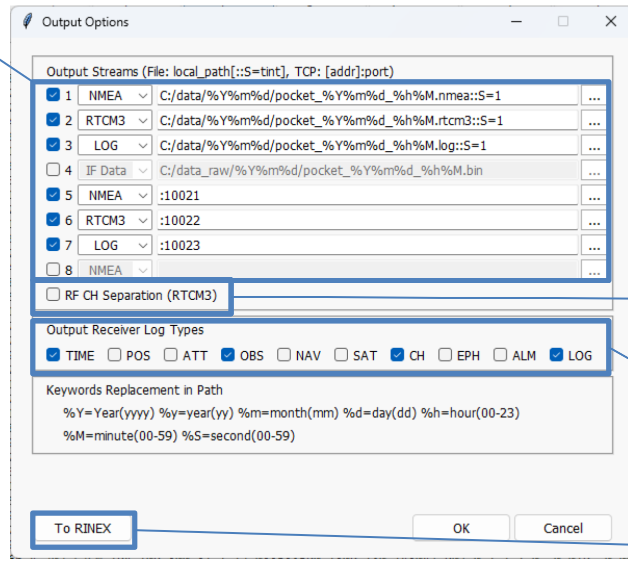
Output Stream 1, 2, 3, ..., 8
ON/OFF, Type and Path

NMEA: PVT Solution *1
(NMEA 0183 GNRMC, GNGGA, GNGSA, GxGSV)

RTCM3: OBS and NAV Data *1 *2
(RTCM 3.4 MSM: MT1077, 1087, 1097, 1117, 1127, 1137 and 1107, with extension and EPH: MT1019, 1020, 1045, 1046, 1044, 1042 and 1041)

LOG: Receiver Log *1 *3 (Receiver Log)

IF Data: IF Data Log *1 *4,
(It is only valid for RF Frontend selected as the input source)



RF CH separation ON/OFF for RTCM3

Receiver Log Type Selections *3

Execute RINEX Converter for RTM3 OBS and NAV data

*1 The output path should be one of the followings.
(1) <local path>: A local file path <local path>. It can contain time keywords replaced by the log start time. Optionally, it can include the file swapping order "::S=<tint>" by the specified interval <tint> (H).
(2) :<port>: TCP server with the port number <port>. The TCP server waits for and accepts the external TCP connection at the port and output data
(3) <addr>:<port>: TCP client with the address <addr> and the port number <port>. The TCP client connects the external TCP server and output data
*2 The output RTCM 3.4 MSM (with extensions) and EPH can be converted to RINEX OBS and NAV by CONVBIN utility included in the Pocket SDR package
*3 Refer to p.16 for the receiver log format. The receiver log can be plotted by the Log Viewer pocket_plot.py in the Pocket SDR package
*4 Refer p.15 for the IF data log format. The output format is automatically selected according to the RF Frontend (RAW8, RAW16, RAW32, CS8 or CS16). The tag file is also generated.

Signal Options Dialog

GNSS System and Signal Selections:

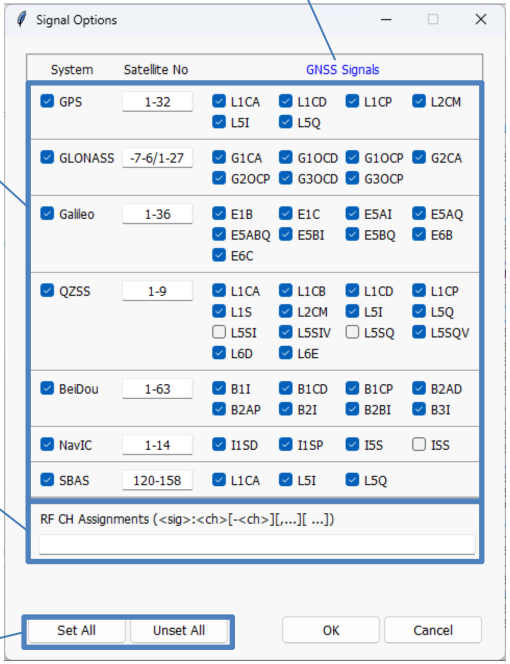
System: GNSS System,
Satellite No: Satellite number or number range (n-m) separated by , *1
GNSS Signals: Signal IDs to be tracked

Link to Detailed GNSS Signal Table

RF CH Assignment Options:

RF CH is automatically assigned for signals. To assign signals to specific RF CH(s) manually, use the option as the format <sig>:<ch_list> separated by spaces, where, <sig> is signal ID and <ch_list> is RF CH number list like 1,3,5-7. Additionally, SRCH:<ch_list> can be used to specify signal search CH(s) to shorten the acquisition time.

Set or unset all systems and signals



Pocket SDR Signal ID Table

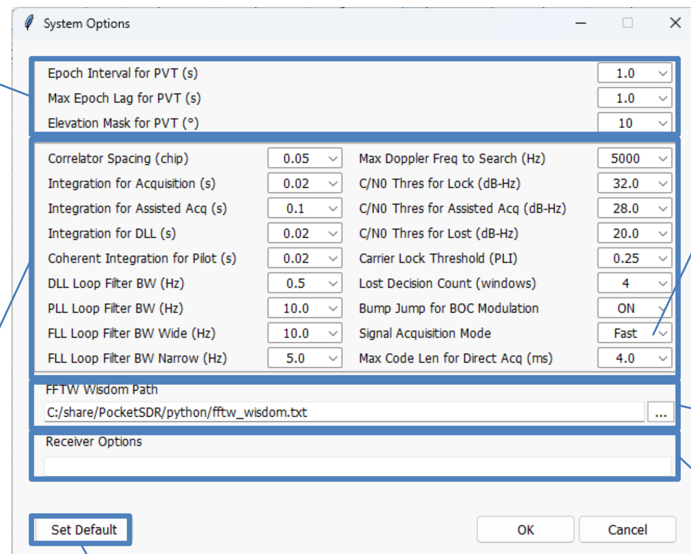
System	Signal	Signal ID	System	Signal	Signal ID
GPS	L1C/A	L1CA	BeiDou	B1I	B1I
	L1C-D	L1CD		B1C-D	B1CD
	L1C-P	L1CP		B1C-P	B1CP
	L2C-M	L2CM		B2a-D	B2AD
	L5-I	L5I		B2a-P	B2AP
	L5-Q	L5Q		B2I	B2I
GLONASS	L1C/A	G1CA		B2b-I	B2BI
	L1OCd	G1OCD		B3I	B3I
	L1OCp	G1OCP	NavIC	L1-SPS-D	I1SD
	L2C/A	G2CA		L1-SPS-P	I1SP
	L2OCp	G2OCP		L5-SPS	I5S
	L3OCd	G3OCD		S-SPS	ISS*3
	L3OCp	G3OCP	SBAS	L1C/A	L1CA
Galileo	E1-B	E1B		L5-I	L5I
	E1-C	E1C		L5-Q	L5Q
	E5a-I	E5AI			
	E5a-Q	E5AQ			
	E5AltBOC	E5ABQ			
	E5b-I	E5BI			
	E5b-Q	E5BQ			
	E6-B	E6B			
	E6-C	E6C			
QZSS	L1C/A	L1CA			
	L1C/B	L1CB			
	L1C-D	L1CD			
	L1C-P	L1CP			
	L1S	L1S			
	L2C-M	L2CM			
	L5-I	L5I			
	L5Q	L5Q			
	L5S-I	L5SI			
		L5SIV*2			
	L5S-Q	L5SQ			
		L5SQV*2			
	L6D	L6D			
	L6E	L6E			

*1 For GPS, Galileo, BeiDou, NavIC and SBAS, these satellite numbers are the same as the signal PRN numbers. For GLONASS, use the frequency channel number (FCN) for FDMA signals and satellite slot number for CDMA signals instead separated by /. For QZSS, which has complicated satellite numbering, use SV IDs (1-5, 8, 9) or non-standard code numbers (6, 10) instead as the satellite numbers.
*2 QZSS L5S verification mode signals
*3 Pocket SDR FE currently does not support S-band signals

System Options Dialog

PVT Options Selection:
Epoch Interval (s), Max Epoch Lag (s) and Elevation angle Mask (deg)

Signal Acquisition and Tracking Options Selection:
Correlation Spacing (chip), Non-coherent Integration Time for Acquisition (s), Non-coherent Integration Time for Assisited Acquisition (s), Non-coherent Integration Time for DLL (s), Coherent Integration Time for Pilot Signal (s), Loop filter bandwidth for DLL, PLL, FLL (wide-band) and FLL (narrow-band) (Hz), Max Doppler Frequency search range (Hz), C/N0 thresholds for signal-lock, Assisted acquisition or signal-lost decision (dB-Hz), Carrier Lock Threshold as PLI (phase lock indicator), Lost Decision Count, Bump-Jump for BOC modulation enabled or disabled, Max code length for direct acquisition (ms).



Set all options default

Signal Acquisition Mode Selection:
Full: All PRN signals are searched.
Fast: Only signals of visible satellites are searched for fast acquisition. The visibility is estimated with CPU time, last or current FIX position, almanac or ephemeris of each satellites. The Doppler also is decided by last or current refence oscillator freq. offset. So, if FE changed, use Full-Search.

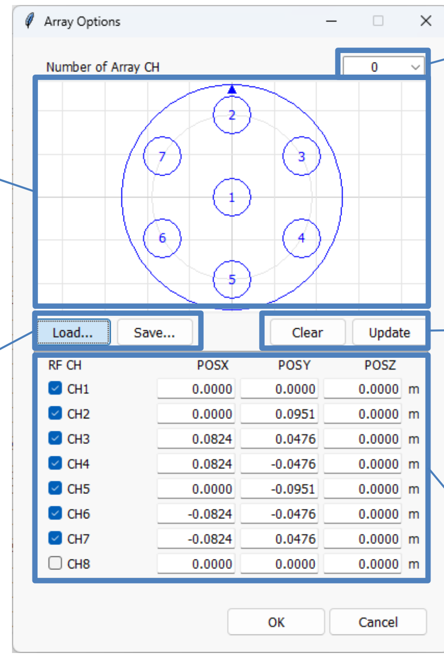
FFTW Wisdom Path:
FFTW wisdom file path to shorten the signal acquisition time. The file can be generated by the utility fftw_wisdom in the Pocket SDR package.

Receiver Specific Options

Array Options Dialog

Array Element Horizontal Plot (Antenna Geometry)

Antenna Geometry Load and Save Buttons

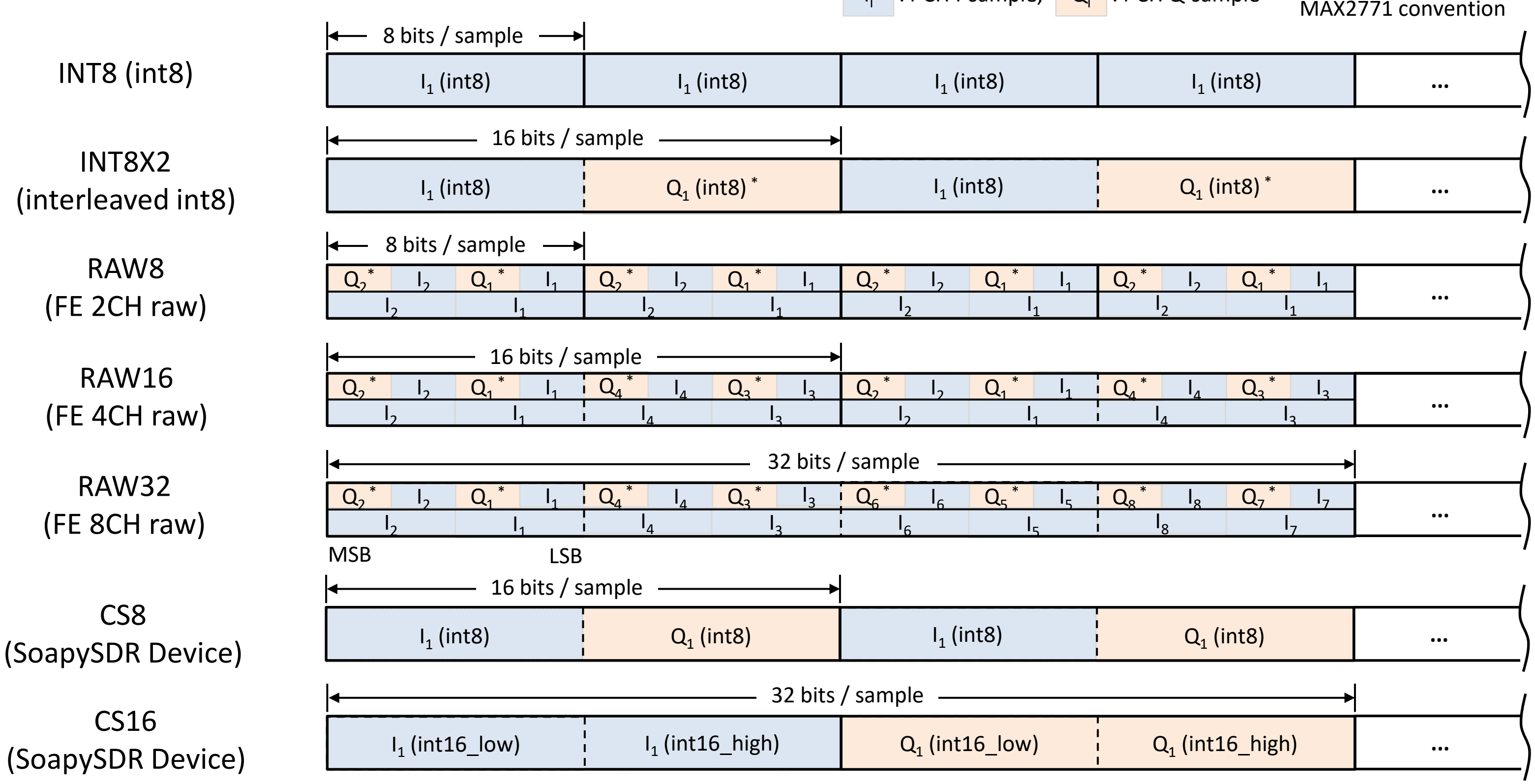


Number of Array CH enabled.

Antenna Geometry Clear and Update Buttons

Enable/Disable and Positions X, Y, Z (m) in body-frame for each Antenna Element.

IF Data File Formats



2 bits IQ-sampling (RAW8/16/32)

bits	00	01	10	11
I	+1	+3	-1	-3
Q *	-1	-3	+1	+3

2 bits I-sampling (RAW8/16/32)

bits	**00	**01	**10	**11
I	+1	+3	-1	-3

3 bits I-sampling (RAW8/16/32)

bits	0000	1000	0001	1001	0010	1010	0011	1011
I	+1	+3	+5	+7	-1	-3	-5	-7

IF Data Tag File (<path>.tag)

```
PROG = Pocket SDR
TIME = 2024/11/01 20:48:00.776      (Start Time GPST)
FMT  = RAW16                        (Format)
F_S  = 24                           (Sampling Freq. MHz)
F_LO = 1568,1237.8,1176.45,1278.75  (LO Freq. MHz)
IQ   = 2,2,2,2                      (Sampling, 1:I, 2:IQ)
BITS = 2,2,2,2                      (Sampling Bits)
```

Receiver Logs

```
$TIME,time,year,mon,day,hour,min,sec,tsys (time info)
$OBS,time,year,month,day,hour,min,sec,sat,code,cn0,pr,cp,dop,lli,fcn (OBS data)
$NAV,time,sat,sig,nerr,size,data (raw NAV data)
$POS,time,year,month,day,hour,min,sec,lat,lon,hgt,Q,ns,stdn,stde,stdu,dtr (position solution)
$SAT,time,year,month,day,hour,min,sec,sat,pvt,obs,cn0,az,el,res (satellite info)
$EPH,time,sat,sig,IODE,IODC,SVA,SVH,Toe,Toc,Ttr,A,e,i0,OMEGA0,omega,M0,delta-n,OMEGAdot,Idot,Crc,Crs,Cuc,Cus,Cic,Cis,Toes,Fit,Af0,Af1,Af2,
  TGD,code,flag (GPS/Galileo/QZSS/BeiDou/NavIC decoded ephemeris)
$EPH,time,sat,sig,tb,fcn,SVH,SVA,age,Toe,Tof,pos-x,pos-y,pos-z,vel-x,vel-y,vel-z,acc-x,acc-y,acc-z,tau-n,gamma-n,delta-tau-n (GLONASS
  decoded ephemeris)
$ALM,time,sat,sig,svh,week,A,e,i0,OMG0,omg,M0,OMGd,toas,f0,f1 (GPS,Galileo,QZSS,BeiDou decoded alamanac)
$ALM,time,sat,sig,svh,taun,lambda,di,eps,omg,tlambda,dT,dTd,frq (GLONASS decoded alamanac)
$CH,time,ch,rfch,sat,sig,prn,lock,cn0,coff,dop,adr,ssync,bsync,fsync,rev,srev,err_phas,err_code,tow_v,tow,week,type,nnav,nerr,nlol,nfec
  (receiver channel info)
$LOG,time,message (error, warning, general info message)
```

```
$TIME
time  receiver time (s)
year  year (2000-2099)
mon   month (1-12)
day   day (1-31)
hour  hour (0-23)
min   minute (0-59)
sec   second (0.000-59.999)
tsys  time system (GPST or UTC)

$OBS
time  receiver time (s)
year,month,day
      obs data day (GPST)
hour,min,sec
      obs data time (GPST)
sat   satellite ID
code  RINEX obs code
cn0   C/N0 (dB-Hz)
pr    pseudorange (m)
cp    carrier phase (cyc)
dop   Doppler frequency (Hz)
lli   loss-of-lock indicator
fcn   frequency channel number
      for GLONASS FDMA signals

$NAV
time  receiver time (s)
sat   satellite ID
sig   signal ID
nerr  number of corrected error
      bits
size  raw NAV data size (bits)
data  raw NAV data HEX dump

$POS
time  receiver time (s)
year,month,day  solution day (GPST)
hour,min,sec  solution time (GPST)
lat   solution latitude
      (deg,+:north,-:south)
lon   solution longitude
      (deg,+:east,-:west)
hgt   solution ellipsoidal height
      (m)
Q     quality flag (=5: single)
ns    number of valid satellites
stdn  solution standard-dev north
      (m)
stde  solution standard-dev east
      (m)
stdu  solution standard-dev up (m)
dtr   receiver clock bias (s)

$SAT
time  receiver time (s)
year,month,day  solution day (GPST)
hour,min,sec  solution time (GPST)
sat   satellite ID
pvt   PVT status (0:not used,1:used)
obs   L1 obs data status
      (0:unavailable,1:available)
cn0   L1 signal C/N0 (dB-Hz)
az    azimuth angle (deg)
el    elavation angle (deg)
res   L1 pseudorange residual (m)

$EPH
time  receiver time (s)
sat   satellite ID
sig   signal ID
...   decoded ephemeris parameters

$ALM
time  receiver time (s)
sat   satellite ID
sig   signal ID
...   decoded almanac parameters

$LOG
time  receiver time (s)
message  error, warning or general info message

$CH
time  receiver time (s)
ch    receiver channel number
rfch  RF channel number
sat   satellite ID
sig   signal ID
prn   PRN number
lock  lock time (s)
cn0   C/N0 (dB-Hz)
coff  code offset (ms)
dop   Doppler frequency (Hz)
adr   accumlated Doppler range (cyc)
ssync secondary code sync flag(0:async,1:sync)
bsync symbol/bit sync flag (0:async,1:sync)
fsync frame sync flag (0:async,1:sync)
rev   primary code polarity (0:normal,1:reversed)
srev  secondary code polarity (0:normal,1:reversed)
err_phas  phase error (cyc)
err_code  code error (10^-6 s)
tow_v  tow valid flag (0:invalid,1:valid,
       2:ambiguity unresolved)
tow   time of week (ms)
week  week number (week)
type  navigation subframe or message type
nnav  navigation subframe/message count
nerr  error subframe/message count
nlol  loss-of-lock count
nfec  number of error corrected (bits)
```